

Mixed Paper

A medium-difficulty assessment focusing on the operational mechanics and types of network firewalls.

SECTION B: Short Answer Questions

Q1. Explain the primary difference between a packet filtering firewall and a stateful inspection firewall. (5 Marks)

SECTION C: Long Descriptive Questions

Q1. Describe how an Application-Level Gateway (Proxy Firewall) operates, and discuss one major advantage and one major disadvantage of using this type of firewall compared to other types. (10 Marks)

INSTRUCTOR ONLY - ANSWER KEY & RUBRIC

Short Questions - Grading Guide

Q1. Ideal Answer:

Packet filtering firewalls inspect packets individually based on headers, while stateful firewalls track the state of active connections. This allows stateful firewalls to make more context-aware security decisions.

Rubric:

Award 2.5 marks for defining packet filtering and 2.5 marks for explaining stateful tracking.

Long Questions - Grading Guide

Q1. Ideal Answer:

An Application-Level Gateway acts as an intermediary that terminates connections to inspect application-layer data. It offers superior security through deep inspection but suffers from high performance overhead.

Rubric:

Award 4 marks for explaining the proxy mechanism, 3 marks for the advantage, and 3 marks for the disadvantage.